

100/1000BASE-T1
SFP Module

USER MANUAL

June 2024

Manual-Version: 0.2 Firmware: 002.002 Product Version: 1.1 or upper

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1 General Information

1.1 Functionality and Features

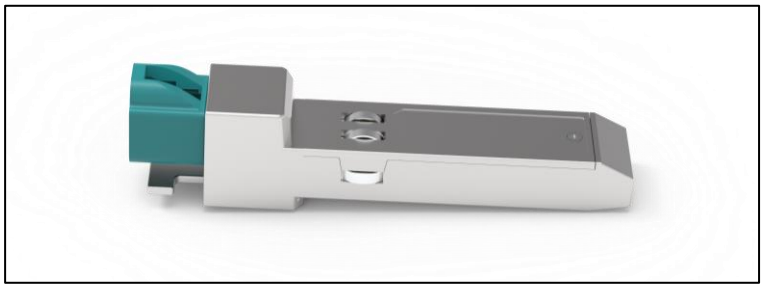


Figure 1-1: 100/1000BASE-T1 SFP

The **Technica Engineering 100/1000BASE-T1 SFP** is a solution to convert between 100/1000BASE-T1 and a host with a 1Gbps SGMII interface.

It fits into a standard SFP (Small Form-factor Pluggable) slot and provides a full-duplex Automotive Ethernet interface with an H-MTD connector that can connect to any automotive DUT.

The device is equipped with one link LED which shows linkup/link-down and also features four DIP switches to define the default configuration at start-up time between different modes:

- Dip switch no. 1: Master / Slave setting
- Dip switch no. 2: 100 / 1000Mbps
- Dip switch no. 3: Reserved
- Dip switch no. 4: Reserved

In compliance with SFF-8472 standard the SFP module contains a memory map with device information at I2C address 0xA0, system integrator can identify the TE-1441 device there and make any custom implementations. Technica is happy to support the integration with any different customer systems.

The module can be used together with other Technica Engineering products, like the Switch-based products, with a seamless integration, for example it can be plugged in Technica Enhanced Ethernet Switch to add additional 100/1000BASE-T1 automotive ethernet ports to our product.

The SFP Module can also be used with any type of hardware offering a suitable SFP slot with a SGMII interface, and additionally also offers an I2C to MDIO gateway feature to any system integrator that needs to extend functionality by directly accessing PHY registers if they desire. On that way, system integrators can fully use all the extended features that the underlying PHY (MARVELL 88Q2221M) has, like TC10 & MACsec.

It also adds a custom implementation to change the DIP switches default status at run-time, so system integrator can change between default PHY configurations offered by Technica without needing to access the complete technical information from the PHY manufacturer*.

**Complete information about PHY 88Q222xM must be acquired under NDA directly from Marvell Technology Inc.*

1.2 Features

- Converts between 100/1000BASE-T1 automotive ethernet and SGMII interfaces.
- Automotive Connector Rosenberger H-MTD
- Marvell 88Q2221M PHY
- 4x Dip switches for configuration
- Fits into a standard SFP slot
- Power requirements: 3.3 Volt DC
- Master/Slave and Speed selection either via small DIP switch or I2C registers
- Supports changes between default configuration modes at run-time without powering down the module through I2C
- Supports internal PHY register access through I2C
- Status LED

1.3 General information

Article number	PT-1441
Voltage requirement:	3.3 Volts DC +/- 0.03 Volts
Power consumption:	Standard SFP Power Level 1 Module (<1W)
Size WxLxH:	68 x 14 x 14 mm
Weight:	0,1 kg
International Protection:	IP 20
Operating temperature:	-40° to +85°Celsius

1.3.1 Links

The user can download the latest documentation for the 100/1000BASE-T1 SFP here: https://portal.technica-engineering.de/myproducts

1.4 Scope of delivery

The delivery includes:

- 1x 100/1000BASE-T1 SFP Module

Cable-set can be ordered separately. Please consult the cable set document with our technicalsales@technica-engineering.de

1.5 Validity

This user manual applies to 100/1000BASE-T1 SFP Module with following product versions:

- Product version 1.1 (using 88Q2221M B1)
- Product version 2.0 and above (using 88Q2221M B2)

2 Safety and warranty

2.1 General operating and safety strategy of TE products

Technica Engineering's products are designed for operation in automotive systems. The applicable limit values adhere to the standard norms for SFP Modules (SFF-8419), automotive onboard power systems correspondingly and can be found in the mentioned norms.

Should Technica Engineering's products be operated in voltage ranges beyond those specified in the norms, which represents a breach of the conditions of operation, then this will void the product warranty and Technica Engineering will assume no liability whatsoever for the results and/or consequences thereof.

This is especially valid whenever the voltage level reaches or exceeds the limits of the low-voltage directive. In this case, damage to the devices cannot be excluded. Due to the manufacturing characteristics of the devices, there is no imminent fire hazard from the device itself, if the devices are being operated in an environment according to the conditions of use. A secondary fire hazard cannot be excluded, should those conditions not be met. Protection against overvoltage cannot be provided in such a breach of the conditions of use.

2.2 General design rules for the power supply of TE products

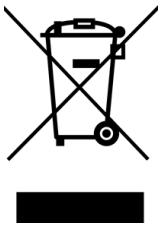
The power supply circuit of Technica Engineering's products is equipped with self-protection components. This automatic function protects the devices against excessive temperature and too high a supply voltage by switching the device off. This automatic switch-off function is independent of any software function.

The root cause of excessive temperature in the power supply circuit can eventually be due to a too high environmental temperature or due to an internal failure of the device. In both cases, the automatic switch-off function will switch off the power supply from the device to avoid further damage.

The protection against too high a supply voltage protects the device even in case of an internal failure of the Technica Engineering device.

2.3 Warranty and safety information

	Before operating the device, read this manual thoroughly and retain it for your reference. The latest documentation for the 100/1000BASE-T1 SFP can be downloaded here: https://portal.technica-engineering.de/myproducts
	Use the device only as described in this manual. Use only in dry conditions. Do not insert any foreign object in the slots/openings of the housing. Do not apply power to a damaged device. The device may only be used by specialists.
	Do not open the device. Otherwise, the warranty will be lost.
	This product is intended for use in automobiles or automotive-like environments. An automotive-like environment includes test setups or test benches in the office, laboratory, and workshop areas. In the test setups, the same environmental conditions apply as in-vehicle electrical systems. Technica Engineering products are not intended to be used as standard IT equipment. The test systems and products from Technica Engineering are designed as customer and application-specific test modules that are only used by specialists for the development and test facilities. When integrating the modules in a vehicle or test set-up, the user must ensure appropriate ventilation or air convection for the device(s). Additionally, the user (Integrator) of this product is the responsible for the external power supply of this module, assuring the appropriate cable protection (fuse). The maximal current load value for the product is described in the user manual. Technica Engineering products must not be considered as a safety element out of context when using safety-critical systems and must be included in the safety assessment when used. The development class in a safety system must be considered with standard QM referred to as ISO26262.

	The device can get hot. Do not cover the device due to fire danger. Do not place the device near highly flammable materials due to fire danger. Do not use the device above the specified operating temperature. The operating temperature is the ambient temperature of the installation space.
	This symbol is only valid in the European Union. If you wish to discard this product, please contact your local authorities or dealer, and ask for the correct method of disposal. Technica Engineering GmbH is registered as a manufacturer of the brand “Technica Engineering” and the device type “Small devices of Information- and Telecommunications- technology for exclusive use in non-private Households”. WEEE Reg. No. DE 20776859

3 Hardware interface

3.1 H-MTD connector

The 100/1000BASE-T1 line is connected by an H-MTD connector.

Name	Picture	Part Number
Rosenberger H-MTD Connector Plug		E6K10A-1CAZ5-Z

Table 1: Parts of Tyco H-MTD Connector

Pinning:

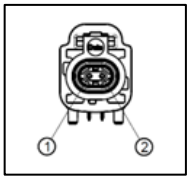


Figure 3-1: H-MTD Connector

Pin	Function
1	Data Line Plus (Positive)
2	Data Line Minus (Negative)

Table 2: H-MTD Connector

This is a shielded connector. The shield in the device is connected to the ground with 10 Ohm and 10 nF.

3.2 SFP socket connector

SFP host connector contact assignment:

Pin	Name	Function	Pin	Name	Function
1	V_{EET}	GND (common with V_{EER})	**11**	V_{EER}	GND (common with V_{EET})
2	TX_FAULT	Transmitter fault (1)	**12**	RD-	SGMII RXD_N
3	TX_DISABLE	Force Link down input (transceiver mute/disable) (2)	**13**	RD+	SGMII RXD_P
4	I2C_SDA	I2C serial interface data line	**14**	V_{EER}	GND (common with V_{EET})
5	I2C_SCK	I2C serial interface clock line	**15**	V_{CCR}	3.3 Volt power supply input
6	MOD_ABS	Module Absent detection (3)	**16**	V_{CCT}	3.3 Volt power supply input
7	RS0	Not connected	**17**	V_{EET}	GND (common with V_{EER})
8	RX_LOS	Loss of Signal Remote Link (4)	**18**	TD+	SGMII TXD_P
9	RS1	Not connected	**19**	TD-	SGMII TXD_N
10	V_{EER}	GND (common with V_{EET})	**20**	V_{EET}	GND (common with V_{EER})

Table 3: Pinning of the connector

General notes

- Labelling as inputs or outputs is from the perspective of the module.

Specific notes

(1) TX_FAULT is open collector output. Should be pulled up with 4.7k – 10kohms on host board to V_{EET}/V_{EER} voltage. A low logic level '0' indicates normal operation in remote 100/1000BASE-T1 link; a high logic level '1' indicates "link error" in remote 100/1000BASE-T1 link.

(2) TX_DISABLE is open collector input with 4.7kohms on module to V_{CCT}/V_{CCR} voltage. A low logic level '0' establishes normal operation on 100/1000BASE-T1 link; a high logic level '1' forces a link down through PHY transceiver disable in remote 100/1000BASE-T1 link. Typically, it will be asserted low or grounded in the host system for normal operation.

(3) MOD_ABS should be pulled up with 4.7k – 10kohms on host board to V_{EET}/V_{EER} voltage. MOD_ABS is connected internally on the module to V_{EET}/V_{EER}, so it pulls line low to notify module is plugged in to the host system.

(4) RX_LOS is open collector output. Should be pulled up with 4.7k – 10kohms on host board to V_{EET}/V_{EER} voltage. A low logic level '0' indicates link up in remote

100/1000BASE-T1 link; a high logic level ‘1’ indicates link down in remote 100/1000BASE-T1 link.

3.3 DIP switches

These DIP switches are physical hardware switches to load default 100/100BASE-T1 configurations on startup time. You can find more information about configuration in CHAPTER 4.1

3.4 Status LED

The 100/1000BASE-T1 SFP has an LED to demonstrate Link and data transfer.

LED	Description
ON	The SFP has a linkup
OFF	There is no link-up
Blinking	Data is being transferred

Table 4 LED Description

4 Start-up & configuration

4.1 DIP switches

The 100/1000BASE-T1 SFP is configured by four DIP switches on the bottom side of the device. Those are intended to define a plug-and-play default start-up configuration of line speed between 100BASE-T1 or 1000BASE-T1 and the mode between slave or master modes.

This configuration can be overridden by accessing I2C register map at any moment, and the module itself will trigger a reconfiguration. For more details of this feature.



Figure 4-1 SFP DIP switches

DIP Switch	Status (physical state of DIP switch)	Description	DIP overriding register bit	Value
1	ON (up) OFF (down)	SFP is set to Master SFP is set to Slave	Bit 7	ON: 0 OFF: 1
2	ON (up) OFF (down)	SFP is set to 1000Mbps SFP is set to 100Mbps	Bit 6	ON: 0 OFF: 1
3	N/A	Reserved	Bit 5	ON: 0 OFF: 1
4	N/A	Reserved	Bit 4	ON: 0 OFF: 1

Table 5: Configuration of DIP Switches

4.2 Startup

After 3.3 Volt power is applied, the SFP module starts up and configures itself according to the DIP switches status. This initial configuration lasts about 100ms. Host system must wait this fixed time window before starting any I2C activity on the bus.

5 I2C interface

5.1 I2C configuration

This chapter includes detailed information about the I2C interface for the SFP module, covering all the functionalities that offers the I2C-MDIO gateway, which is intended for reading and writing PHY's registers. Also detailed information is added about the dip switches overriding feature and how to use it properly.

The module operates with fSCL up to 400kHz without requiring clock stretching. The module processor listens as a slave on the 7-bit address 0x50 (8-bit address 0xA0) following SFF-8472 MSA (Multi-source Agreement) specification for I2C access to module operating parameters.

b1010 000X = 0xA0

The read operation requires the register address and the number of bytes to be read. When the acknowledgment is received from the SFP module, the data bytes are serially clocked out.

Example of reading Register 0 of the SFP Memory map:

HOST	START	I2C addr							R/W		Register addr								
		MSB				LSB			Write	ACK	MSB				LSB				ACK
		1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

START	I2C addr							R/W		Data word 1									STOP
	MSB				LSB			Read	ACK	Data								NACK	
	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	1	1	1	

Figure 5-1 Read operation of register 0

5.2 Standard SFP modules I2C memory map

In compliance with SFF-8472 standard the SFP module contains a memory map with device information at I2C address 0xA0.

It is recommended to the system integrator that product is identified by using **Vendor PN** field, as it is a fixed field that will never change in newer hardware/software revisions of the product.

It is recommended to the system integrator to recognize product version by using field **Vendor Rev**, which will always consist in a major version and a minor version in

"X.Y"format. Upon reading the product version system integrator can decide to use features that could be available only on some specific product revision.

Other fields are populated in compliance with SFF-8472 standard and any system integrator has freedom to use them for any integration purposes.

Data Address	Field Name	Data	Comment
0	Identifier	03	SFP
1	Ext. Identifier	04	
2	Connector	80	Vendor-specific H-MTD
3-10	Transceiver	00 00 00 08 00 00 00 00	
11	Encoding	06	
12	Signaling Rate, Nominal	67	
13	Rate Identifier	00	
14	Length (SMF,km) or Copper Cable Attenuation	00	
15	Length (SMF) or Copper Cable Attenuation	00	
16	Length (50 um, OM2)	03	
17	Length (62.5 um, OM1)	01	
18	Length (OM4 or copper cable)	0F	
19	Length (OM3) or Cable length, additional	0F	
20-35	Vendor name	54 65 63 68 6E 69 63 61 20 45 6E 67 2E 20 20 20	"Technica Eng. "
36	Transceiver	00	
37-39	Vendor OUI	38 2A 19	Technica OUI
40-55	Vendor PN	54 45 31 34 34 31 20 20 20 20 20 20 20 20 20 20	"TE1441 " It must be used for integration purposes to detect product as it will not change during product lifecycle.
56-59	Vendor rev	31 2E 31 20	"1.1 " It must be used for integration purposes to detect available features. 1.1 à 88Q2221M B1 2.x à 88Q2221M B2
60-61	Wavelength	00 00	
62	Fibre Channel Speed 2	00	
63	CC_BASE	xx	
64-65	Options	00 1A	
66	Signaling Rate, max	00	

67	Signaling Rate, min	00	
68-83	Vendor SN	31 34 34 31 31 31 30 30 32 34 30 32 30 30 30 31	Different for each produced device following Technica SN format: 1441110024020001
84-91	Date code	32 33 31 31 31 35 20 20	"231115 " 84-85: ASCII code, two low order digits of year. (00 = 2000). 86-87: ASCII code, digits of month (01 = Jan through 12 = Dec) 88-89: ASCII code, day of month (01-31) 90-91: ASCII code, vendor specific lot code, may be blank
92	Diagnostic Monitoring Type	00	
93	Enhanced Options	00	
94	SFF-8472 Compliance	03	
95	CC_EXT	xx	
96-127	Vendor Specific	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
128-255	Reserved	FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF	

Table 6 SFP Memory map

5.3 Gateway I2C-MDIO

The SFP module includes a Gateway I2C-MDIO custom feature that enables communication between an I2C master and the PHY within the module. This allows

the I2C master to read from and write to PHY registers by utilizing the microcontroller as a bridge between the host and the PHY.

The I2C-MDIO gateway listens as a slave on the 7-bit address 0x40 (8-bit address 0x80), and when accessing the gateway for read/write operations, a 16-bit register address is required. The most significant byte (MSB) of the register address, value 0xFF, serves as the gateway addressing indicator, while the least significant byte (LSB) indicates the gateway register to be accessed. This 16-bit I2C registers addressing needs to be considered by the system integrator.

Essentially, the gateway operates as a 16-bit I2C memory. Here is an example of its usage:

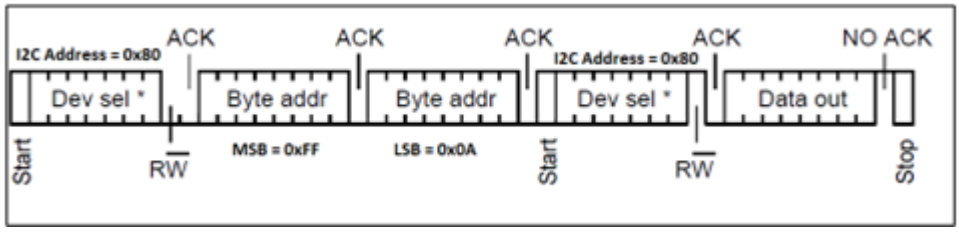


Figure 5-2 SFP I2C-MDIO gateway read register transaction

To utilize the I2C-MDIO gateway, the I2C address should be set to 0x80, followed by writing to the gateway matching register (0xFF) and then the gateway operation register (refer to Table 4). Upon detecting the gateway matching register (0xFF), the Gateway I2C-MDIO is activated. Subsequently, one of the provided register addresses can be used to specify the register for either reading or writing.

Register Address	Register Name
0x00	MDIO Device number
0x01	PHY Register address MSB
0x02	PHY Register address LSB
0x03	PHY Register Operation Status
0x04	PHY Register data MSB
0x05	PHY Register data LSB
0x0A	DIP Switches overriding

Table 7 I2C-MDIO Gateway register addresses

These register addresses provide access to various functionalities and information related to the I2C-MDIO Gateway. By reading from or writing to these registers, the host can communicate with the PHY and perform MDIO operations as needed.

The individual registers are further described in detail below:

5.3.1 I2C-MDIO Gateway Register Description

- **MDIO Device number**: Defines the device number of the register to access.

- **PHY Register address MSB**: Defines the Most significant byte of the MDIO register to access.
- **PHY Register address LSB**: Defines the Less Significant Byte of the MDIO register to access.
- **PHY Register Operation Status**: Used to control of get status of MDIO read/write operations.
    - **Bit 0** – MDIO Read Start Condition Flag (Read/Write): This flag is set by the master after specifying the first 3 bytes of the register (Devices number, Register address MSB, and Register address LSB) when a read operation is triggered.
    - **Bit 1** – Read In Progress Flag (Read): This flag is set by the slave during the reading operation.
    - **Bit 2** – Read Operation Done Flag (Read): This flag is set by the I2C slave when the MDIO read operation is finished. This flag can be read after triggering the read to ensure that the data has finished reading.
    - **Bit 3** – MDIO Write Start Condition Flag (Read/Write): This flag is set by the master after specifying the first 3 bytes of the register (Devices number, Register address MSB, and Register address LSB) when a write operation is triggered.
    - **Bit 4** – MDIO Write In Progress Flag (Read): This flag is set by the slave during the writing operation.
    - **Bit 5** – MDIO Write Operation Done Flag (Read): This flag is set by the I2C slave when the MDIO write operation is finished. This flag can be read after writing the PHY register data LSB to ensure that the data has finished writing.
    - **Bit 6** – Reserved
    - **Bit 7** – Reserved
- **PHY Register data MSB:** When a reading operation, this register contains the MSB of the PHY MDIO data register. When a write operation, this register contains the MSB to write in the PHY MDIO register.
- **PHY Register data LSB:** When a reading operation, this register contains the LSB of the PHY MDIO data register. When a write operation, this register contains the LSB to write in the PHY MDIO register.
- **DIP switches overriding:** When the SFP module is powered on loads the default configuration got from sampling the physical DIP switches. Depending on the meaning of each of the switches in ON/OFF states the SFP module can start in different initial configurations. This feature allows a host system integrator to dynamically change the default startup configuration by just sending a new DIP switches configuration and trigger a PHY reset to load that new configuration. Useful as an additional abstraction layer to avoid needing to know PHY MDIO memory map to make dynamic changes to the SFP module behavior.

5.3.2 MDIO write operation example

To make a MDIO write operation, it is needed a combination of an I2C write transaction to perform the MDIO write operation on the bus and an I2C read transaction to query the Write Operation Done Flag to assure the completion of the operation.

To write data to the I2C-MDIO gateway control registers, the host system I2C master can write individually each of the control registers or write all of them in a bulk write operation to write 6 register bytes in a single I2C transaction.

To perform a MDIO write operation, the I2C transaction should start with and I2C Start Condition plus the I2C address of the I2C-MDIO gateway with appropriate write bit set (0x80), followed by two bytes that indicate the gateway access register (0xFF) and the initial register which will be accessed, Device Number (0x00) plus six data bytes:

- MDIO Device Number
- MDIO Register Address MSB
- MDIO Register Address LSB
- Register Operation Status with the Write Start Condition flag (0x08)
- Two data bytes to be written to the MDIO register.

HOST	START	I2C addr							R/W		Gateway addr									GW - Register 0								
		MSB						LSB	Write	ACK	MSB							LSB	ACK	MSB							LSB	ACK
		1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0

Dev Addr								ACK	Reg Addr MSB								ACK	Reg Addr LSB								ACK	Reg Op. Status								ACK
MSB							LSB		MSB							LSB		MSB							LSB		MSB							LSB	
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	0	0	0

Reg 4 Content								ACK	Reg 5 Content								ACK	NACK	STOP
MSB							LSB		MSB							LSB			
0	0	0	1	0	0	0	0	0	0	1	0	1	1	1	0	0	0	1	

Figure 5-3 I2C write transaction of all I2C-MDIO gateway registers to trigger a MDIO write operation

If the MDIO write operation is successful, the Write Operation Done Flag will be finally set by the device and can be queried by the host system I2C master in a I2C read transaction.

The I2C transaction should start with and I2C Start Condition plus the I2C address of the I2C-MDIO gateway with appropriate write bit set (0x80), followed by two bytes that indicate the gateway access register (0xFF) and the initial register which will be accessed, Register Operation Status (0x03). That will be followed by a Repeated Start condition plus the I2C address of the I2C-MDIO gateway with appropriate read bit set (0x81) followed by the data in the desired register in the I2C slave that will be read by the I2C master.

If the previous MDIO write operation is successful, the Write Operation Done Flag will be set by the module and the value (0x20) will be finally returned in this register. The host system can implement a polling mechanism to assure MDIO transaction has been performed in the physical bus.

HOST	START	I2C addr							R/W		Gateway addr									GW - Register 3								
		MSB						LSB	Write	ACK	MSB							LSB	ACK	MSB							LSB	ACK
		1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	1	1	0

START	I2C addr								R/W	Reg 3 Content										STOP
	MSB							LSB	Read	MSB							LSB		ACK	NACK	
	1	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	1	

Figure 5-4 I2C Read operation of register 0x03 to check Write Operation Done Flag

5.3.3 MDIO read operation example

To make a MDIO read operation, it is needed a combination of an I2C write transaction to trigger the MDIO read operation on the bus and two I2C read transactions, first one to query the Read Operation Done Flag to assure the completion of the operation, and the second one to retrieve the data read from the MDIO register itself.

Phy Addr	R/W	GW Addr	Reg Addr	Dev addr	Reg addr MSB	Reg addr LSB	Reg Op Status	Reg Data MSB	Reg Data LSB
0x80	W	0xFF	0x00	0x01	0x00	0x00	0x01	-	-
Wait for 50ms, or check I2C gateway register 3 (Op status) until bit 2 is set (Read Operation Done Flag)									
0x80	W	0xFF	0x00	0x04					
0x80	R	-	-	-	-	-	-	0x20	0x5C

Figure 5-5 I2C frames representing a read operation

To perform a MDIO read operation, the I2C transaction should start with and I2C Start Condition plus the I2C address of the I2C-MDIO gateway with appropriate write bit set (0x80), followed by two bytes that indicate the gateway access register (0xFF) and the initial register which will be accessed, Device Number (0x00) plus four data bytes

- MDIO Device Number
- MDIO Register Address MSB
- MDIO Register Address LSB
- Register Operation Status with the Read Start Condition flag (0x01)

HOST	START	I2C addr							R/W		Gateway addr									GW - Register 0								
		MSB						LSB	Write	ACK	MSB							LSB	ACK	MSB							LSB	ACK
		1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0

Device number									Register MSB								Register LSB									Reg Op. Status									STOP
MSB							LSB	ACK	MSB							LSB	ACK	MSB							LSB	ACK	MSB							LSB	ACK	
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	

Figure 5-6 I2C write transaction of all I2C-MDIO gateway registers to trigger a MDIO read operation

If the MDIO read operation is successful, the Read Operation Done Flag will be finally set by the device and can be queried by the host system I2C master in an I2C read transaction.

The I2C transaction should start with and I2C Start Condition plus the I2C address of the I2C-MDIO gateway with appropriate write bit set (0x80), followed by two bytes that indicate the gateway access register (0xFF) and the initial register which will be accessed, Register Operation Status (0x03). That will be followed by a Repeated Start condition plus the I2C address of the I2C-MDIO gateway with appropriate read bit set (0x81) followed by the data in the desired register in the I2C slave that will be read by the I2C master.

If the previous MDIO read operation is successful, the Read Operation Done Flag will be set by the module and the value (0x04) will be finally returned in this register. The host system can implement a polling mechanism to assure MDIO transaction has been performed in the physical bus.

HOST	START	I2C addr							R/W		Gateway addr									GW - Register 3								
		MSB						LSB	Write	ACK	MSB							LSB	ACK	MSB							LSB	ACK
		1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	1	1	0

START	I2C addr							R/W		Reg 3 Content								NACK	STOP
	MSB						LSB	Read	ACK	MSB							LSB		
	1	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	1	

Figure 5-7 I2C Read operation of register 0x03 to check Read Operation Done Flag

Finally, as the MDIO read transaction has been finished the host system I2C Master can then proceed to retrieve the data read from the PHY through another I2C read transaction. The data is available in the PHY MDIO Data Registers, located at I2C register addresses 0x04 and 0x05. The master must perform a third read operation of 2 bytes from register address 0x04 to retrieve the data.

HOST	START	I2C addr							R/W		Gateway addr									GW - Register 4								
		MSB						LSB	Write	ACK	MSB							LSB	ACK	MSB							LSB	ACK
		1	0	0	0	0	0	0	1	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0

START	I2C addr							R/W		Reg 4 Content								Reg 5 Content								NACK	STOP
	MSB						LSB	Read	ACK	MSB							LSB	ACK	MSB							LSB		
	1	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	1	1	1	0	0	1	

Figure 5-8 I2C Read transaction of registers 0x04 and 0x05 to retrieve the data read from MDIO register

5.3.4 DIP Switches overriding

The DIP switches overriding is a functionality intended to change the configuration loaded through the sampling of physical DIP switches condition during SFP+ module start-up, by using I2C transactions. Through the reading and writing of the register DIP Switches overriding register, the host system is able to read the physical DIP switches configuration and change it to a completely new configuration. The value of the DIP switches is represented by a byte, and the DIP switches disposition is stored in the

MSB of that byte. The DIP switches are represented in active low logic (ON:0 and OFF:1)

Data word1							
MSB				LSB			
1	0	0	0	x	x	x	x

Figure 5-9 Virtual DIP switches register map interpretation

If the SFP+ module detects DIP switches overriding coming from the host system through and I2C transaction, it triggers a PHY reset and a full configuration with the new parameters.

The feature is useful in case the host system integrator does not want to know about underlying PHY specific registers configuration, either by choice or because unavailability of the PHY register map information*, allowing the possibility to make dynamic changes to the module configuration in a higher abstraction layer level.

*Complete information about PHY 88Q2221M must be acquired under NDA directly from Marvell Technology Inc.

That dynamic changes are volatile and not permanent on the device, it means that if a power off – on cycle is triggered, for example if the SFP+ module is extracted and inserted again, the device will sample again the physical DIP switches status and would load that initial configuration again, and not to the last configuration sent through I2C bus.

DIP SWITCHES REGISTER I2C READ EXAMPLE

The following frame shows an I2C read transaction of the register 0x0A, with the 16-bit addressing and the repeated start condition to get the data from the register.

HOST	START	I2C addr							R/W		Gateway addr								GW - Register 10								
		MSB						LSB	Write	ACK	MSB							LSB	ACK	MSB							LSB	ACK
		1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	1	0	0

START	I2C addr							R/W		Reg 10 Content								NACK	STOP
	MSB						LSB	Read	ACK	MSB							LSB		
	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	1	

Figure 5-10 Dip switches read operation

DIP SWITCHES REGISTER I2C WRITE EXAMPLE

The following frame shows an I2C write transaction of the register 0x0A, sending the value of 0x00 to the DIP switches register.

HOST	START	I2C addr							R/W		Gateway addr								GW - Register 10								Data word1									
		MSB						LSB	Write	ACK	MSB							LSB	ACK	MSB							LSB	ACK	MSB							LSB	NACK	STOP
		1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	1	

Figure 5-11 Dip switches write operation

6. Electrical specifications

Technica Engineering's TE-1441 100/1000BASE-T1 SFP module is compliant with the SFP Electrical MSA - SFF-8431 specification for improved pluggable form factor modules.

6 Mechanical specifications

Technica Engineering's TE-1441 100/1000BASE-T1 SFP module is compliant with the SFP Mechanical MSA - SFF-8432 specification for improved pluggable form factor modules.
The SFP module also has an opener lid latch for easy extraction of the modules.
Finally it also contains an automotive ethernet -H-MTD connector from Rosenberger.

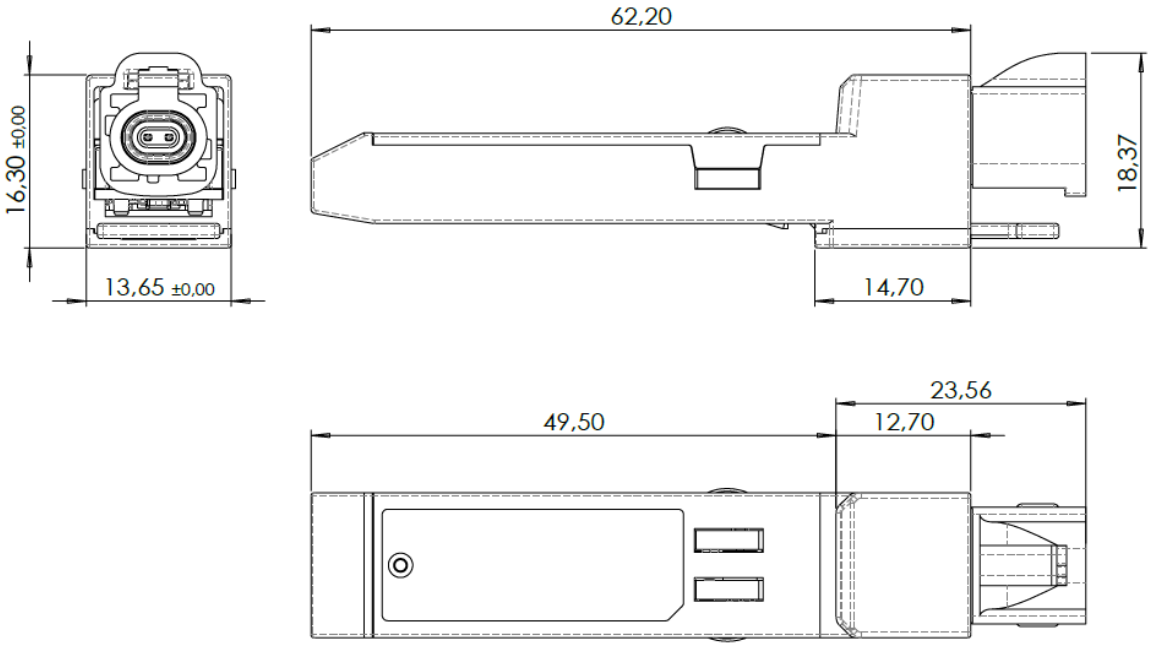


Figure 6-1 Mechanical drawings

7 Disclaimer and additional information

- The 1000BASE-T1 Port is optimized for automotive use cases. The maximum line length for each 1000BASE-T1 segment is limited to 10 meters.
- Since devices 88Q2221M B1 are still qualified as engineering samples prototypes, some issues while using SFP modules containing them can appear due to B1 silicon limitations.*
- Functional hardware and/or software upgrades for older silicon samples can't be guaranteed by Technica Engineering.

*Further specific information about PHY 88Q2221M silicon differences must be acquired under NDA directly from Marvell Technology Inc.

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9 Changelog

Version	Chapter	Description	Date
0.1	All	First release	Feb 2024
0.2	All	Major redesign	June 2024

10 Contact

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